Gaurav Kunwar

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Research Interests

Interpretable and trustworthy machine learning; efficient, quantized inference for Transformers; deep learning for biomedical signals and images (ECG, EEG, EHR); computer vision; graph neural networks.

Education

M.Tech, Computer Science and Engineering, National Institute of Technology (NIT) Silchar, India 2023–2025
Thesis: *Sugarcane Leaf Disease Detection and Quality Assessment*, supervised by Dr. Aparajita Dutta; supported by a CSIR (Govt. of India) ASPIRE grant.
Coursework: Machine Learning, Artificial Intelligence, Linear Optimization, Advanced Data Structures, Advanced DBMS, High-Performance Computing.

B.Tech, Computer Science and Engineering, GRD Institute of Management and Technology, Dehradun, India 2019–2023
Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, Artificial Intelligence, Machine Learning.

Academic Appointment

Assistant Professor (Grade III), DIT University, Dehradun, India Aug 2025–present

- Teach C Programming, R Programming and Operating Systems to about 120 undergraduates per semester.
- Design and evaluate examinations and assignments; mentor student projects; faculty coordinator for the university farewell event; helped set up a student disciplinary committee.

Research Experience

Low-precision Transformers and generative models for biosignals Lead author, 2026

- Trained a dual-resolution ResNet–Transformer masked autoencoder for 12-lead PTB-XL ECG with simulated MXINT8 arithmetic in every Transformer linear layer: Pearson r 0.990, PRD 14.19% at 500 Hz, 0.87 M parameters, est. 1.61 MiB packed vs. 3.36 MiB FP32.
- Compared class-conditional WGAN-GP and DDPM generators for minority-class MIT-BIH beats; augmentation raised five-class macro-F1 of a 1D-CNN from 0.662 to 0.750.
- Built a per-subject k-means pipeline on 139,803 ear-EEG epochs from 128 subjects; a label audit found every stage label set to -1, so the study was reframed as unsupervised discovery and a leakage-resistant 88/20/20 subject split was released.

Interpretable Vision Transformers for plant disease M.Tech thesis, NIT Silchar

- Built a ViT pipeline with Squeeze-and-Excitation blocks and Position Attention Networks; compared with Inception V3 and MobileNet.
- Merged four public datasets into a 2,401-image healthy vs. red-rot set; ViT reached F1 96.25% vs. 93.41% for MobileNet under k-fold cross-validation; used Integrated Gradients to test whether predictions rest on lesions or background.

Graph neural networks for EHR mortality prediction With Prof. Prateek Srivastava

- Built cosine-kNN patient-similarity graphs from MIMIC-IV; trained GCN and multi-head GAT models with SMOTE and weighted loss; interpreted predictions with SHAP and t-SNE.

Publications

Accepted and published

[1] **G. Kunwar**, A. Dutta. "Sugarcane Leaf Diseases Detection and Quality Assessment through Image Processing with Deep Learning Algorithms." *4th Int. Conf. on Networks and Cryptology (NetCrypt 2026)*, JNU, New Delhi. Accepted, oral presentation.

[2] P. Maisnam, **G. Kunwar**. "Machine Learning-Based Benchmarking of Structural Variant Detection Using Sequencing Microbes Data." *IC2E3-2026*, NIT Uttarakhand. Accepted and presented.

[3] **G. Kunwar**, P. Srivastava. "Explainable Disease Risk Prediction from Electronic Health Records Using Graph Convolutional Network and SHAP." *AUTOCOM-2026*. Accepted.

[4] **G. Kunwar**, R. Jha. "Prediction of Air Pollution using Deep Learning based LSTM and CNN Model." *IEEE ICAECT 2023*. doi:10.1109/ICAECT57570.2023.10118248

[5] R. Jha, **G. Kunwar**. "Predicting the Patient's Severity Using Machine Learning Applied to Lungs MRI Images." *IEEE ISCON 2023*. doi:10.1109/ISCON57294.2023.10111986

[6] R. Jha, **G. Kunwar**. "Machine Learning based URL Analysis for Phishing Detection." *IEEE ISCON 2023*. doi:10.1109/ISCON57294.2023.10112057

Under review

[7] **G. Kunwar**, D. Sengupta, Medhavi, S. Yadav. "Comprehensive Hardware-Aware Quantization of Microscaling Integer Biosignal Transformers for Edge Diagnostics." *ICMC 2027*, IIT Guwahati (Springer LNNS).

[8] **G. Kunwar**, D. Sengupta, Medhavi, P. Kumar. "Comparing Conditional GAN and Diffusion Models for Synthetic ECG Generation and Minority-Class Augmentation." *ICMC 2027*.

[9] **G. Kunwar**, D. Sengupta, P. Kumar, Medhavi. "Subject-Calibrated Unsupervised Spectral-State Discovery from Ear-EEG: A Reproducible Pilot Study with Explicit Label-Audit Reporting." *ICMC 2027*.

[10] R. K. Rai, S. Parui, M. Mahapatra, **G. Kunwar**, R. P. Naik, D. K. Singh. "NeuroSciML-Net: Physics-Informed EEG Learning for Cognitive State Classification During Hare Krishna Mahamantra Meditation and Kirtan Kriya." *ICVGIP 2026* (ACM).

[11] A. Acharya, D. Sengupta, P. Jain, Medhavi, S. Pattnaik, A. Nath, A. K. Parida, **G. Kunwar**. "Spatiotemporal Physics-Aware Autoencoder for Joint Estimation of Underwater Radiance and Attenuation." *IEEE RAIT 2027*, IIT (ISM) Dhanbad.

[12] D. Sengupta, Medhavi, P. Jain, A. Acharya, S. Pattnaik, A. Nath, A. K. Parida, **G. Kunwar**. "Hardware Security of Deep Neural Network Models and Accelerators: Threats, Vulnerabilities, and Countermeasures." *IEEE RAIT 2027*.

[13] "Human Cognitive Behavior and Brain State Analysis in Different Physiological Conditions using EEG Signal: A Systematic Review." *Computer Science Review* (Elsevier).

Service

Reviewer, *IEEE Transactions on Consumer Electronics* (2026).

Skills and Languages

Programming: Python, C, C++, Java, R, Julia, JavaScript, HTML/CSS
ML and scientific computing: TensorFlow, scikit-learn, NumPy, pandas, FEniCS, DeepXDE, NVIDIA Modulus
Methods: Vision Transformers, CNNs, LSTMs, GCN/GAT, GANs, diffusion models, quantization-aware training, SHAP, Integrated Gradients, physics-informed neural networks
Tools: Git, MySQL, Flask, LaTeX, draw.io
Languages: Hindi (native); English (fluent; medium of instruction for B.Tech and M.Tech)

References

Dr. Aparajita Dutta, M.Tech Thesis Supervisor, Dept. of CSE, NIT Silchar — aparajita.dutta@cse.nits.ac.in
Dr. Ripon Patgiri, Dept. of CSE, NIT Silchar — ripon@cse.nits.ac.in
Dr. Ramanujam E, Dept. of CSE, NIT Puducherry — ramanujam.e@nitpy.ac.in